Lesson notes · for you

Cells, rows and formulas

Data engineering · Level 2 · Lesson 1

Time
15 min

Materials
A phone or computer with a spreadsheet app, or squared paper; Your phone with Darsi, for the Python steps

Learning goals

- Find your way round a spreadsheet: cells and ranges.
- Use SUM, AVERAGE and COUNTIF.
- Work out a total and an average in Python.

Activity: Formulas by hand

1. Grid (3 min): Draw a grid with columns A to C and rows 1 to 6. Put a heading in row 1 and five marks in column B.
2. Address (3 min): A partner calls out addresses like B3; you say the value in that cell.
3. Formula (5 min): Write =SUM(B2:B6) and =AVERAGE(B2:B6) under the table and work them out.
4. Change (4 min): Change one mark. What changes in the total and the average? On a computer, it would happen by itself.

Use made-up marks. If you have a spreadsheet app, try the same formulas on it.

Discussion

- What does the colon in B2:B9 mean?
- Why does a formula beat a calculator?
- Why can’t SUM add “14 / 17”?

Check yourself

- Write a formula for the average of cells C2 to C10.

Try it this week
Make a table of something you track for a week, like hours of sleep. Add a total and an average with formulas. Then change one number, and watch them update.

Worksheet

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Name _____ Class _____ Date _____

1 **Why is a formula better than typing the total yourself?**

◯ It hides the marks ◯ It makes the numbers bigger ◯ It updates when a mark changes

2 **Match each piece of a spreadsheet to what it means.**

B2	• • • •	Every cell from B2 to B9
B2:B9	• • • •	How many of those cells say 6A
=AVERAGE(B2:B9)	• • • •	The cell in column B, row 2
=COUNTIF(C2:C9,"6A")	• • • •	The mean of those cells

3 **What does this program print?**

```
marks = [14, 9, 17, 12]
print(sum(marks), len(marks))
```

◯ 52 4 ◯ 14 9 17 12 ◯ 4 52

4 **Fill the gaps to work out the average mark.**

```
marks = [14, 9, 17, 12]
average = _____(marks) / _____(marks)
print(average)
```

Word bank: `max` `print` `sum` `len`

5 **Zainab's class has eight marks. Fix the program so it prints the total, then the average.**

```
marks = [14, 9, 17, 12, 15, 11, 13, 13]
total = 0
print(total)
print(total / len(marks))
```

Write your program:

What would you like to keep track of in a spreadsheet? Which formula would you use?

Check your answers

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1 **It updates when a mark changes**

Change a mark and the formula works out the total and the average again by itself, with no new adding-up mistakes.

2 **B2 — The cell in column B, row 2**
B2:B9 — Every cell from B2 to B9
=AVERAGE(B2:B9) — The mean of those cells
=COUNTIF(C2:C9,"6A") — How many of those cells say 6A

An address names one cell, a range names a block, and formulas such as AVERAGE and COUNTIF work on a range.

3 **52 4**

sum(marks) adds the four marks, 52, and len(marks) counts them, 4.

4

```
marks = [14, 9, 17, 12]
average = sum(marks) / len(marks)
print(average)
```

The average is the total divided by how many: sum(marks) / len(marks) = 52 / 4 = 13.0.

5

```
marks = [14, 9, 17, 12, 15, 11, 13, 13]
total = sum(marks)
print(total)
print(total / len(marks))
```

total = sum(marks) adds the eight marks: 104.